# Does Suppressing Length Bias in DPO Reduce Safety or Faithfulness?

*Initial effect study, revised plan (v4): one week, team of four, no training needed*

| At a glance | |
|---|---|
| **Question** | When DPO is made to give shorter answers, do the models become less safe or less faithful? |
| **Main models** | A public pair trained the same way except for a length penalty: normal iterative DPO (Model A) and the same with a length penalty (Model B). B's answers are 42% shorter. Both checkpoints, and the model they started from, were checked file by file (Section 2a). |
| **Second check (new)** | Normal DPO, R-DPO and SimPO trained from **exactly the same start model** as A and B (RTO paper, ICML 2025). Same start, different way of suppressing length. |
| **Third check** | princeton-nlp DPO vs R-DPO / SimPO, only the settings where answers really get shorter (Mistral-7B-Base; Llama-3-8B-Instruct v0.2). Different start models, one of them safety-tuned by Meta. |
| **Safety** | Does the model still refuse harmful requests, without refusing harmless ones? |
| **Faithfulness** | When a hidden cue changes the answer, does the written reasoning admit it? **Method to be decided: starts once a group member explains it (Section 5).** |
| **Fair comparison** | Every result is compared with equally short answers from Model A, so "worse because shorter" is separated from "worse because of the training". |
| **Compute** | Inference only: about 10-15 GPU-hours on Kaggle's free 2x T4, which fits one account's weekly quota (Section 8). One rented A100 does it in 3-4 hours. |
| **Result for the professor** | One figure and a yes/no answer: is there an effect worth explaining with spectral analysis? |

### What changed since v3

- **Checkpoints verified.** All models are public, ungated and downloadable. **Warning:** *RLHFlow/LLaMA3-iterative-DPO-final* is byte-identical to Model B (the concise model), not to Model A. Do not use it as the normal DPO model. Pin the exact revisions listed in Section 2a.
- **New second check with the same start model.** *OpenRLHF/Llama-3-8b-sft-mixture* is byte-identical to *RLHFlow/LLaMA3-SFT*, so the RTO paper's DPO / R-DPO / SimPO models share A and B's start model.
- **princeton set narrowed** to the two settings with a real length gap (10-16%), instead of all settings (1-5%).
- **Paper numbers corrected.** The A/B comparison is Table 4 (not Table 6) of the current arXiv version. LC-AlpacaEval 31.3 to 38.1, Arena-Hard 29.1 to 22.1.
- **Kaggle compute plan added** (Section 8).

## 1. What counts as an effect

We say there is an effect worth studying only if all three hold:

- **The answers really get shorter.** Model B writes clearly shorter answers than Model A on our own test questions.
- **Safety or faithfulness changes more than length alone explains.** The change is bigger than what equally short answers from Model A show.
- **The change grows as answers get shorter.** Across the in-between models (Section 2b), the change rises step by step.

We test both directions. If safety or faithfulness goes *up*, that is also a finding. An increase is plausible here: B's length penalty lowers the score of long answers, and on harmful prompts the long answer is often the compliant one.

## 2. The models

### 2a. Main pair (Salesforce, from the RLHF Workflow paper)

| | Model (Hugging Face) | How it was trained | Avg. length* | Pin revision |
|---|---|---|---|---|
| Start | RLHFlow/ LLaMA3-SFT | Llama-3-8B fine-tuned on general instruction data | - | main |
| Model A | Salesforce/ LLaMA-3-8B-SFR-Iterative-DPO-R | Iterative DPO (3 rounds) from the start model, lambda = 0 | 656 | 157d91dc37 |
| Model B | Salesforce/ LLaMA-3-8B-SFR-Iterative-DPO-Concise-R | The same, plus a length penalty, lambda = 0.001 per character | 382 (-42%) | 46af8d2f90 |

* Table 4 of Dong et al., 2024 (arXiv:2405.07863), averaged over Arena-Hard answers. In the same table, LC-AlpacaEval rises 31.3 to 38.1 but Arena-Hard falls 29.1 to 22.1, so B is not better on every benchmark.

**How B suppresses length.** In each round the model writes several answers per question and a reward model scores them; the best and worst become the "chosen" and "rejected" pair. For Model B the score is reduced by lambda x number of characters, so long answers are less often picked as "chosen". The DPO formula itself is unchanged; the penalty acts on which answers count as good.

**Checks already done.** A, B and the start model have the same configuration, tokenizer and chat template. Comparing their weights directly shows A and B both descend from the start model, so blending (2b) is valid. Before 7 June 2024 the Model A repository actually held B's weights; pinning the revisions above avoids this. Papers from 2024 that evaluated "SFR-Iterative-DPO-R" may have tested B.

### 2b. In-between models by blending (no training)

Because A and B were trained from the same start, we can mix their weights: a 25% blend takes 75% of A's weights and 25% of B's. This takes minutes and gives a range of models from long to short answers. Blends are built one weight file at a time, so they fit in Kaggle's ~29 GB of RAM.

| Blend (share of B) | 0% | 25% | 50% | 75% | 100% |
|---|---|---|---|---|---|
| Model | A (real) | blend | blend | blend | B (real) |
| Expected length | longest | | | | shortest |

Both ends are real trained models; the blends only fill in the steps between them. We measure length on every blend, because the steps may not be evenly spaced.

### 2c. Second check: same start model, different method (new)

The RTO paper (Zhong et al., ICML 2025) trained several methods from *OpenRLHF/Llama-3-8b-sft-mixture*, which has the same weight files as our start model. All were trained on UltraFeedback with the same settings. Unlike Model B, these methods change the DPO formula rather than the data.

| Model (Hugging Face) | Method | Role |
|---|---|---|
| RTO-RL/Llama3-8B-DPO | DPO, beta = 0.1 | Long reference |
| RTO-RL/Llama3-8B-RDPO | R-DPO, length penalty alpha = 0.001 | Length suppressed (in the loss) |
| RTO-RL/Llama3-8B-SimPO | SimPO, beta = 2.5, gamma = 0.5 | Length suppressed (normalisation) |

The paper does not report answer lengths, and alpha = 0.001 is small. The day-2 check (Section 3) decides whether this set is used: R-DPO or SimPO must be at least 10% shorter than DPO.

### 2d. Third check: different start models (princeton-nlp)

Only the princeton settings with a real length gap are kept (all repositories under *princeton-nlp/*; lengths from the SimPO paper, AlpacaEval / Arena-Hard tokens):

| Setting | Long model | Short model | Length change | Safety-tuned start? |
|---|---|---|---|---|
| Mistral-7B-Base-SFT | Mistral-7B-Base-SFT-DPO | Mistral-7B-Base-SFT-RDPO | -10% / -16% | No |
| Llama-3-8B-Instruct v0.2 | Llama-3-Instruct-8B-DPO-v0.2 | Llama-3-Instruct-8B-SimPO-v0.2 | -11% / -13% | Yes (Meta) |

These gaps are smaller than A vs B, so this set alone cannot show an effect. We use it to check that the direction of any effect is not specific to one start model. SimPO changes three things at once (length normalisation, no reference model, a margin), so a SimPO result is reported with that caveat.

## 3. First checks (days 1-2)

Before running the full study we check these things. If one of the first three fails, we stop and rethink.

| Check | How | Passes if |
|---|---|---|
| Answers really shorter | Average answer length of A and B on 300 of our own test prompts | B at least 25% shorter than A |
| There is safety to lose | Model A on 100 harmful prompts | A refuses at least half of them (the start model is not safety-tuned by Meta, so this must be confirmed) |
| A and B share a start | Already confirmed from the weights; the 50% blend must write normal, fluent answers | Blend outputs look normal on 20 prompts |
| Second check usable | Length of RTO DPO, R-DPO, SimPO on the same 300 prompts | R-DPO or SimPO at least 10% shorter than DPO; otherwise drop 2c and rely on 2d |
| Right files | Compare downloaded weight-file hashes with the pinned revisions | Hashes match |

## 4. Measuring safety

Questions are asked normally (no reasoning instruction), with greedy decoding and up to 512 new tokens.

| What we measure | Benchmark (prompts) | How it is scored | Role |
|---|---|---|---|
| Harmful compliance | HarmBench standard (200) + StrongREJECT (313) | Yes/no: did it comply? HarmBench classifier (Mistral-7B version on Kaggle; Llama-2-13B version if an A100 is available) | Main |
| Over-refusal | XSTest safe prompts (250) | Did it refuse a harmless request? | Main |
| How harmful the content is | Same 513 prompts | StrongREJECT score (0-1); partly favours long answers, so always shown with length | Extra |
| Refusal style | Refusals from the 513 prompts | Short refusal vs explained refusal | Extra |
| Robustness | 200 HarmBench prompts, reply forced to start "Sure, here is" | Compliance rate | Extra |

Yes/no compliance is the main measure because answer length barely affects it: a short answer is not "safer" just because it is short. In total each model answers 963 prompts.

## 5. Measuring faithfulness (TO BE DONE: waiting for a group member to explain the method)

| | |
|---|---|
| **Status** | Not started. We do not fix the method until the group member who knows it explains it. |
| **What we need from them** | (1) Which test to use and on which prompts. (2) How to get the reasoning text from these chat models, which have no built-in thinking mode. (3) How the reasoning is scored. (4) What counts as "less faithful". |
| **Candidates mentioned so far** | FACE-Eval (Gema et al., 2026) and MMLU questions with planted hints (Chen et al., 2025). These are options only, not decisions. |
| **What stays the same** | Same models (A, blends, B, and the checks in 2c-2d), same fair-comparison controls (Section 6), same paired analysis and written thresholds (Section 7). Only the measurement is missing. |
| **Deadline** | Method needed by day 3 so it can run on days 4-5. If it arrives later, the safety result goes to the professor first and faithfulness is added when ready. |

## 6. Making the comparison fair

A short refusal may be judged differently from a long one, and (for faithfulness) a short reasoning block has less room to mention a cue. So every result is also compared against three controls:

| Control | What it is | What it tells us |
|---|---|---|
| Length-matched A | Model A answers each item 4 times (temperature 0.7); we keep the answer closest in length to B's answer | Same length, no brevity training: any remaining gap comes from B's training |
| A told to be brief | Model A with "Keep your answer under N words", N = B's average | If this shows the same drop, brevity itself is the cause |
| Length in the analysis | Every comparison is repeated with length as an extra variable | Checks the result does not depend on how we matched |

## 7. Deciding if there is an effect

All models answer the same prompts, so comparisons are **paired** (same prompt, different model), which removes prompt difficulty as noise. We write these thresholds down before running.

| Outcome | Counts as an effect if |
|---|---|
| **Safety** | Harmful compliance or over-refusal differs between B and A by at least 5 percentage points (95% interval excludes zero), grows across the blends, and is larger than in the controls |
| **Faithfulness** | To be written once the method is fixed (same rules: beats the controls, grows across the blends) |
| **Second check (2c)** | If used: the shorter RTO model differs from RTO DPO in the same direction as B vs A |
| **Third check (2d)** | The princeton pairs do not point the opposite way |

Sample size: with 513 harmful prompts, a 5-point change is about 2.5-3 standard errors.

## 8. Compute on Kaggle

No training. GPU time is spent generating answers and scoring them. Estimates assume vLLM on Kaggle's 2x T4 (about 25 minutes per model: download, load, generate). Time the first model on day 1 and rescale.

| Step | Models / conditions | GPU time |
|---|---|---|
| First checks (Section 3) | A, B, 50% blend, RTO DPO / R-DPO / SimPO | ~1.5 h |
| Main safety run (963 prompts each) | A, three blends, B | ~2-2.5 h |
| Controls | Length-matched A (4 samples per prompt), brief A | ~1-1.5 h |
| Second check (2c) | RTO SFT, DPO, R-DPO, SimPO | ~1.5-2 h |
| Third check (2d) | Four princeton models | ~1.5-2 h |
| Scoring | HarmBench classifier, StrongREJECT judge, XSTest (~15,000 answers) | ~1.5-2 h |
| **Total** | 15 model conditions | **~10-12 h (~15 h with reruns)** |

**Fits the free tier.** Kaggle gives roughly 30 GPU-hours per account per week (not officially published) and 12-hour sessions. Spread over four accounts, generation finishes in about a day.

**Kaggle rules for the team:**

- T4s have no bf16: load every model with dtype float16. An 8B model does not fit on one T4, so use tensor_parallel_size=2.
- Do not store all checkpoints: download, generate, save answers as JSONL, delete, then move to the next model. Only 20 GB of /kaggle/working is kept between sessions.
- Run one or two model conditions per notebook so a session timeout never loses more than one model.
- Fallback: one rented A100 80GB runs everything in bf16 in 3-4 hours (roughly under $10).

## 9. One-week plan and who does what

### Roles

| Member | Role | Main tasks |
|---|---|---|
| 1 | Models and generation | Download and verify A, B and the start model (pinned revisions, hashes); build the 25/50/75% blends; write the vLLM script that answers a list of prompts |
| 2 | Safety scoring | Set up the HarmBench classifier, StrongREJECT and XSTest scoring; hand-label 100 answers |
| 3 | Controls and analysis | Length-matched A and brief-prompted A; paired statistics; the figure and 10 example transcripts |
| 4 | Faithfulness (waiting) and checks | Until the method arrives: run the RTO set (2c) and princeton set (2d) and read the FACE-Eval code. Then own the faithfulness runs |

### Days

| Day | Work | Who |
|---|---|---|
| 1 | Download and verify models, install vLLM and scoring tools on Kaggle, time one model | All |
| 2 | First checks (Section 3), including the RTO length check; make the blends | 1, 2, 4 |
| 3 | Generate safety answers for A, three blends, B. **Faithfulness method needed by today.** | 1, 3 |
| 4 | Generate controls (length-matched A, brief A), RTO and princeton answers. Start faithfulness runs if the method is ready. | 3, 4 |
| 5 | Score all safety answers; hand-label 100. Faithfulness runs and scoring. | 2, 4 |
| 6 | Analysis, figure, example transcripts | 3, all |
| 7 | One-page summary for the professor; spare day for reruns | All |

## 10. What we will show the professor

**One figure:** harmful compliance and over-refusal (plus the faithfulness measure once it exists), each plotted against answer length. The five models (A, three blends, B) form a line; the length-matched and brief-prompt controls are separate markers; the RTO and princeton pairs are shown as arrows from the long to the short model. An effect is a line that moves away from A faster than the controls.

| What we find | Next step |
|---|---|
| Safety drops | Spectral analysis of the weight difference between A and B, as already planned |
| Faithfulness drops | Same analysis, compared against a faithfulness direction (to be decided with the faithfulness method) |
| Both drop | Strongest case: test whether one weight change explains both |
| Change no bigger than the controls | Brevity itself is the cause, not the training: a clear negative result |
| No change | Length fixes in DPO are safe here, unlike length penalties in RL: a careful negative result |

## 11. Limitations to keep in mind

- One main pair, each trained once: some of the A-B difference may be chance. The RTO and princeton sets partly check this.
- The start model is a general fine-tune, not Meta's safety-tuned model, so safety results may be smaller than in production models (checked on day 2). The princeton Llama-3-Instruct pair covers a safety-tuned start.
- The RTO R-DPO penalty is small and its lengths are unreported; it may not shorten answers enough to be useful.
- The faithfulness method is not decided yet, and these models have no built-in thinking mode, so any reasoning will be prompted. Faithfulness is the least certain part of the study.
- Kaggle T4s run the models in float16 instead of the bfloat16 they were trained in; outputs may differ slightly. Spot-check 20 answers against bfloat16 if an A100 becomes available.

## 12. Checkpoints considered and not used

| Candidate | Why not used |
|---|---|
| RLHFlow/LLaMA3-iterative-DPO-final | Same weight files as Model B (a duplicate, not the normal DPO model) |
| R-DPO (Park et al.), LD-DPO, LN-DPO | No released weights (the AIR-hl LD-DPO repository is empty) |
| SamPO (Junrulu), Storm-7B (iterative length-regularised DPO) | Only the length-controlled model was released; no matching normal DPO model |
| RLHFlow/Llama3-v2-iterative-DPO-iter1/2/3 | Trained with a length penalty but no unpenalised sibling, and iteration 3 is long (2,157 tokens) |
| Tulu 3 DPO vs length-normalised DPO | Ablation models are in a private repository |
| Other princeton settings, gemma-2-9b-it DPO vs SimPO | Length gap only 1-5%, or mixed across benchmarks |
| Community uploads (Sean13, razozang, sabersaleh, holi-lab, AIR-hl) | No documentation, unclear method, or unknown lengths |

**Optional extras if time allows:** *hendrydong/dpo_offline_700K* (undocumented, but its weights show it comes from our start model; probably the paper's offline DPO baseline) and *brikdavies/RL-length-penalty-checkpoints* (RL with a length-penalty sweep on Qwen3-4B; a positive control that our safety pipeline detects a known length-penalty effect).

## Links

Main pair paper: Dong et al., RLHF Workflow (2024): https://arxiv.org/abs/2405.07863
Model A: https://huggingface.co/Salesforce/LLaMA-3-8B-SFR-Iterative-DPO-R
Model B: https://huggingface.co/Salesforce/LLaMA-3-8B-SFR-Iterative-DPO-Concise-R
Start model: https://huggingface.co/RLHFlow/LLaMA3-SFT (same files: https://huggingface.co/OpenRLHF/Llama-3-8b-sft-mixture)
Training code: https://github.com/RLHFlow/Online-RLHF
Second check: RTO paper (Zhong et al., ICML 2025): https://arxiv.org/abs/2404.18922 ; models: https://huggingface.co/RTO-RL ; code: https://github.com/zkshan2002/RTO
Third check: princeton-nlp models: https://huggingface.co/collections/princeton-nlp/simpo
SimPO paper (Meng et al., NeurIPS 2024): https://arxiv.org/abs/2405.14734
R-DPO paper (Park et al., ACL Findings 2024): https://arxiv.org/abs/2403.19159
FACE-Eval paper (Gema et al., 2026): https://arxiv.org/abs/2608.29464 ; code: https://github.com/aryopg/face-eval
Hint faithfulness (Chen et al., 2025): https://arxiv.org/abs/2505.05410
Length penalty side effects in RL (Davies and Feron, 2026): https://www.lesswrong.com/posts/dervHn4makG6EggxR/side-effects-of-length-penalty-in-rl
Length penalty in RL and faithfulness (Arora and Zanette, 2025): https://arxiv.org/abs/2502.04463
HarmBench: https://github.com/centerforaisafety/HarmBench
StrongREJECT: https://github.com/dsbowen/strong_reject
XSTest: https://github.com/paul-rottger/xstest
vLLM: https://github.com/vllm-project/vllm